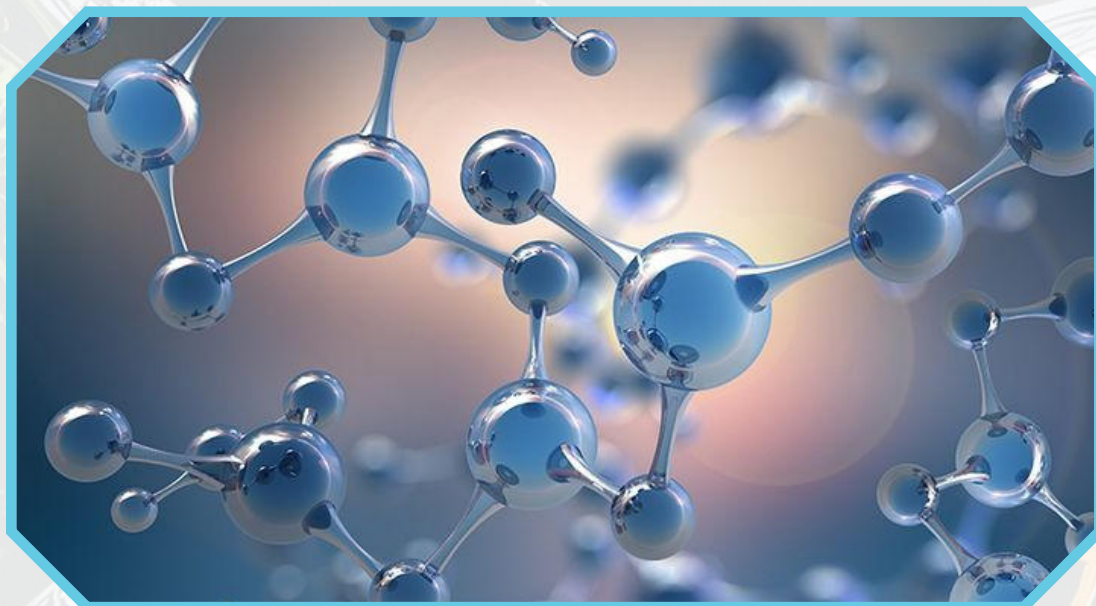




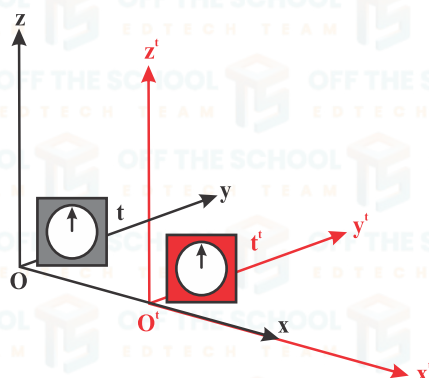
UNIT 24

RELATIVITY



Frame of References:

A rigid framework (usually x , y , and z -axes) relative to which the position and motion of an object can be measured is called frame of reference.



Relative Motion:

Relative motion is a concept that describes the motion of an object in relation to another observer or frame of reference. This viewpoint recognizes that motion is not absolute but depends on the observer's point of view.

Types of Frames of References:

There are two types of frames of reference, Inertial frames of reference and non-inertial frames of reference.

• Criteria:

- Inertial frame of Reference
- Non-Inertial frame of Reference

• **Acceleration:** Inertial frames are not accelerating, while non-inertial frames are.

• **Newton's Laws:** Newton's laws of motion apply directly in inertial frames. In non-inertial frames, additional forces (called fictitious or pseudo-forces) must be introduced to explain the observed motion.

Examples:

- **Inertial frame:** A car moving at a constant speed on a straight highway.
- **Non-inertial frame:** A car accelerating or turning a corner, a rotating merry-go-round.

Special Theory of Relativity:

Special theory of relativity explains how to interpret motion between different inertial frames of reference.

In other words, it deals with the problems in which one frame of reference moves with a constant linear velocity relative to another frame of reference.

The special theory of relativity is based on two essential assumptions, commonly known as postulates.

Postulate 1 (Principle of Relativity):

The laws of Physics have the same form in all inertial frames of reference.

Postulate 2 (Constancy of the speed of light):

The speed of light in vacuum has the same value, $c = 3 \times 10^8 \text{ m/s}$, in all inertial reference frames, regardless of the velocity of the observer or the velocity of the source emitting the light.

Space and Time are Relative:

When the speed of light is constant, the time and space become relative concepts and are no longer absolute according to Newton.

Consequences of Special Theory of Relativity:

In Relativity, there is no such thing as an absolute length or absolute time interval.

Here are some of the most important consequences of special relativity:

Relativity of Simultaneity:

Boy A's Perspective (on the Train):

- Boy A is sitting inside the moving train. The train is moving at 200 km/h, and Boy A fires an arrow at the target. From Boy A's perspective, the arrow moves towards the target at the normal speed, and he perceives the lightning striking at both ends of the train simultaneously.
- This means that, in Boy A's reference frame, both lightning strikes reach him at the same time. Since Boy A is moving along with the train, the light from both ends reaches him simultaneously, regardless of the train's speed.

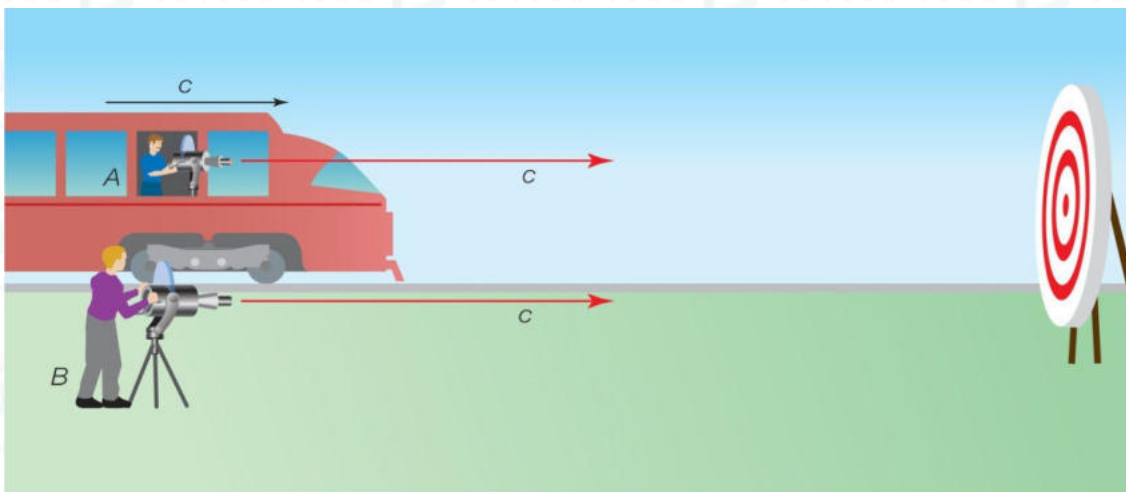
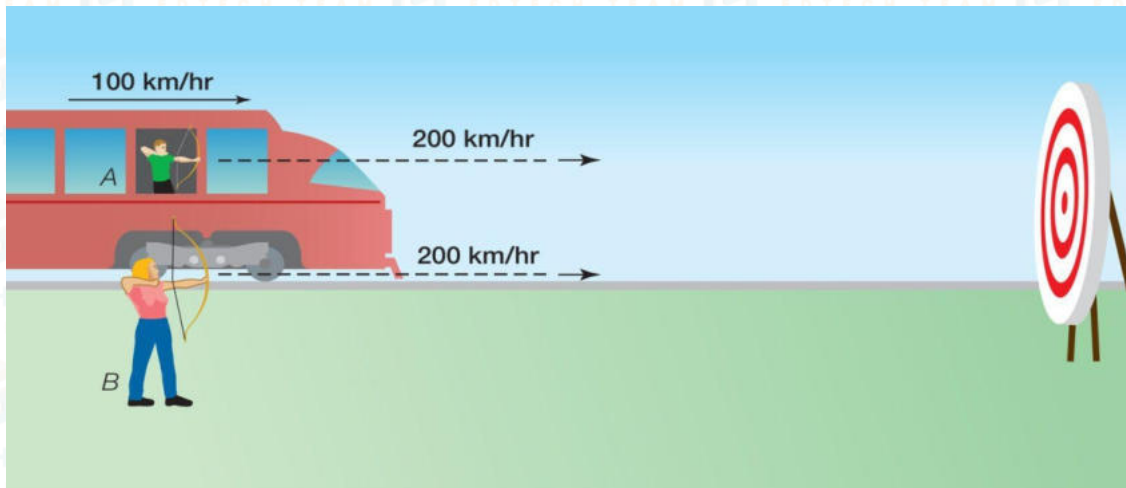
Boy B's Perspective (Outside the Train):

- Boy B is stationary on the ground, some distance from the train. He sees the train moving at 100 km/h relative to him.
- From Boy B's perspective, as the train is moving toward the lightning strike at the front of the train and away from the lightning strike at the back, the light from the front strikes Boy B first (because the train is moving toward the front), while the light from the back takes longer to reach him.
- Therefore, Boy B perceives the lightning strikes as happening **at different times**, even though both strikes occurred at the same moment for Boy A (who is in the moving train).

Key Point of Relativity of Simultaneity:

In **Special Relativity**, simultaneity is not absolute. Events that appear simultaneous in one reference frame (Boy A inside the moving train) may not appear simultaneous in another reference frame (Boy B standing on the ground). This is because the speed of light is constant for all observers, regardless of their motion, and their perception of time and space is relative to their state of motion.

This situation shows that simultaneity depends on the observer's frame of reference. While Boy A (inside the train) perceives the lightning strikes as simultaneous, Boy B (stationary on the ground) perceives them as happening at different times due to the relative motion of the train.



Time Dilation:

The time interval, between two events occurring at a given point in the moving frame S' , appears to be longer to the observer in the stationary frame S . This effect is called time dilation.

Let Δt_0 be the proper time measured by a clock that is at rest. The relative time t measured in another frame of reference is given by:

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Length Contraction:

Let L_0 be the length of a rod when the rod is stationary. If there is relative motion at speed v between an observer at rest and the rod along the length of the rod, then the observer will calculate a relativistic length L given by:

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Mass Variation:

Let m_0 be the rest mass of an object. If the object is moving at speed v then its relativistic mass m will be given by:

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Mass Energy Relationship: For an object to travel at light:

According to Einstein's special theory of relativity, mass and energy are interchangeable. An object's mass m and the equivalent energy E , are related by:

$$E_0 = mc^2$$

If the object is moving, it has additional energy in the form of kinetic energy K .

The total energy E is the sum of its mass energy and its kinetic energy:

$$E = E_0 + K = mc^2 + K$$

where γ is the Lorentz factor for the object's motion.

E is the total relativistic energy of the object.

From above equations, relative kinetic energy can be calculated as:

$$K = E - E_0 = E - mc^2 = \gamma mc^2 - mc^2$$

$$K = mc^2 (\gamma - 1)$$

General Theory of Relativity:

Albert Einstein's general theory of relativity is a new way to explain gravity.

According to Newton, tossed balls curve because of a force of gravity.

According to Einstein, tossed balls and light don't curve because of any force, but because the space-time in which they travel is curved.

General relativity describes how gravity is caused by the curvature of spacetime.

The general theory of relativity is based on the following two postulates.

Postulate I:

The laws of physics may be expressed in equations having the same form in all frames of reference, regardless of their states of motion.

Thus, the general theory covers uniform as well as accelerated motion.

Hence it is able to describe gravitational phenomena.

Postulate II (Principle of equivalence):

There is no way to spot the difference between gravity and acceleration for an observer in a closed laboratory.

General relativity has a number of consequences:

- Einstein's general theory of relativity states that time is a fourth dimension adding to the three dimensions of space. Einstein called this four-dimensional geometry as space-time.
- The general theory of relativity also predicted light coming from a strong gravitational field would have its wavelength shifted toward lower wavelengths, called a red-shift.
- The theory also predicted that when gravity becomes great enough, it would produce objects called black holes. Black holes are objects whose gravity is so massive that light cannot escape from the surface at all. Since no light can escape, such objects would appear black.
- Light is sent as it passes through curved space-time. This can cause distant objects to appear distorted or magnified. Gravitational lensing has been used to study distant galaxies and other objects.

- **The expansion of the universe:** General relativity predicts that the universe is expanding. This has been confirmed by astronomical observations.

Gravity as Space Time Continuum:

According to General Theory of Relativity, when you put mass in space-time, it bends the geometry of space-time.

According to this theory, gravity is not just a force between masses, as described by Newtonian physics, but rather a manifestation of the curvature of space-time caused by mass and energy.

Here's a breakdown of what this concept entails:

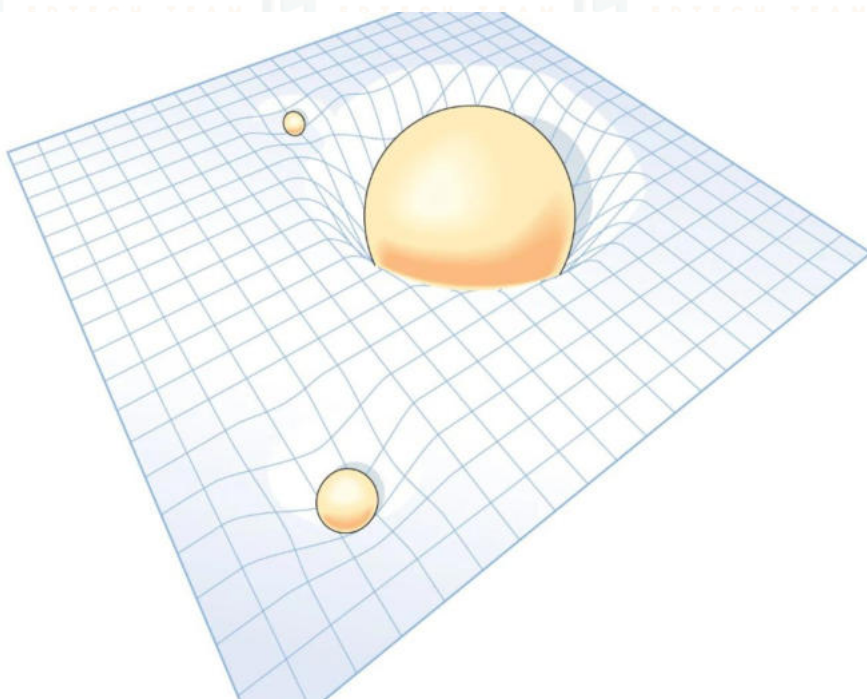
Space and time Continuum

Space and time are unified into a single, four-dimensional entity known as space-time.

In the absence of gravity or significant mass, space time is flat.

However, the presence of mass and energy warps or curves the fabric of space-time.

This curvature is what we perceive as the force of gravity.



Curvature of Space-time:

Massive objects, such as stars, planets, and galaxies, curve the space-time around them. The greater the mass or energy density, the greater the curvature of space-time.

- Objects move along paths dictated by the curvature of space-time, which we perceive as being influenced by gravity. For example, Earth orbits the Sun because the Sun's mass curves the space-time around it, causing Earth to follow a curved path.

Effects of Gravity:

- In the presence of gravity, objects follow the shortest possible paths, known as geodesics, through the curved space-time. These paths are not necessarily straight.
- Lines in the traditional sense but are instead curved trajectories determined by the gravitational field.
- The strength of gravity is determined by the curvature of space-time.

The steeper the curvature, the stronger the gravitational force experienced by nearby objects.

Physical Interpretation:

Gravity is thus interpreted as the result of objects moving through the curved space-time created by mass and energy. Massive objects "warp" the space time around them, causing other objects to move in response to this curvature.

This concept provides a unified explanation for both the gravitational force and the motion of objects in space, incorporating both space and time into a single framework.